



WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **Renew Coil Engineering & Supply Co., Ltd.**

WPS No. **RNC-WPS-016** Date **29-Mar-13** Revision No. **0**

Supporting PQR No.(s) **RNC-PQR-016** Date **29-Mar-13** Type(s) **Manual**

Welding Process(es) **Gas Tungsten Arc Welding (GTAW)** (Automatic, Manual, Machine, or Semi-Auto.)

JOINT (QW-402)

Joint Design **Single V**

Backing (Yes) ☐ (No) ☒ X

Backing Material (Type) **N/A**

☐ Metal ☐ Nonfusing Metal
☐ Nonmetallic ☐ Other

Notes

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. **4** Group No. **1** to P-No. **4** Group No. **1**

Material Specification Type and Grade **A335 P11, F182-F11, A234 WP11** TO **A335 P11, F182-F11, A234 WP11**

Chem. Analysis and Mech. Prop.

Thickness Range :

Base Metal : Groove **1.6 mm - 16.36 mm.** Fillet **All**

Pipe Dia. Range : Groove **All** Fillet **All**

Other

FILLER METALS (QW-404)

	GTAW	
Spec. No. (SFA)	5.28	
AWS No. (Class)	ER80S-G	
F-No.	6	
A-No.	3	
Size of Filler Metals	Ø 2.4 ~ 3.2mm.	
Deposited Weld Metals	8.18 mm.	
Thickness Range :		
Groove	16.36 mm.	
Fillet	All	
Electrode-Flux (Class)	-	
Flux Trade Name	-	
Consumable Insert	-	
Other		

POSITION (QW-405)

Position(s) of Groove **ALL**

Welding Progression Up ☒ X Down ☐

Position(s) of Fillet **ALL**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **704°C~746°C**

Time Range **Holding Time 2 Hour.**

Notes **Heating Rate=220°C Max./Cooling Rate=250°C Max**

PREHEAT (QW-406)

Preheat Temp. Min. **200°C~250°C**

Interpass Temp. Max. **300°C**

Preheat Maintenance **N/A**

Notes **Resistance heater**

GAS (QW-408)

Percent Composition		
Gas(es)	(Mixture)	Flow Rate
Shielding	Argon	99.99%
Trailing	-	-
Backing	-	-



WELDING PROCEDURE SPECIFICATION (WPS)

WPS No. RNC-WPS-016 Rev. 0

ELECTRICAL CHARACTERISTICS (QW-409)

Current ☐ AC ☒ DC Polarity EN
Amps (Range) 80-150 Volts (Range) 8-13
Tungsten Electrode Size and Type Ø 2.4 mm, Thoriated (EWTh-2)
(Pure Tungsten, 2% Thoriated, etc.)
Mode of Metal Transfer for GMAW N/A
(Spray arc, short circuiting arc, etc.)
Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

String or Weave Bead BOTH APPLICATIONS
Orifice or Gas Cup Size 8.0 mm - 19.0 mm
Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING
Method of Back Gouging N/A
Oscillation N/A
Contact Tube to Work Distance N/A
Multiple or Single Pass (per side) MULTIPASS
Multiple or Single Electrodes SINGLE ELECTRODE
Travel Speed (Range) 6-9 Cm/Minute
Peening N/A
Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER-80S-G	2.4	DC EN	120-150	8-12	4-6	
2nd	GTAW	ER-80S-G	2.4	DC EN	120-150	10-13	8-1	
Other(s)	GTAW	ER-80S-G	2.4	DC EN	80-130	7-11	6-9	

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirement of ASME-IX, 2007 latest Revision.

Prepared by: Mr. Chatchawan S. 11/04/13
Name Chatchawan S. Signed [Signature] Date 11/04/13
Reviewed by: Phuchin Seehomchai 11/04/13
Name Phuchin Seehomchai Signed [Signature] Date 11/04/13
Approved by: Phuchin Seehomchai 11/04/13
Name Phuchin Seehomchai Signed [Signature] Date 11/04/13



Procedure Qualification Record (PQR)

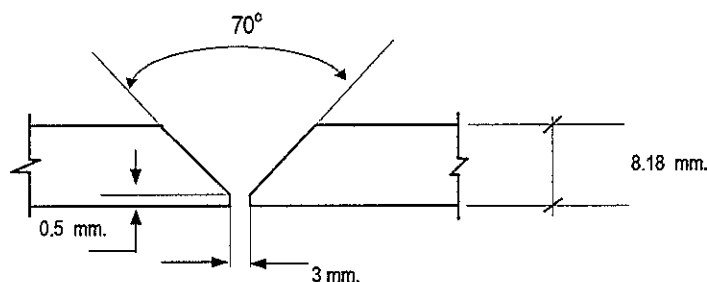
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : RNC-PQR-003

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-016	Date :	29 March 2013
WPS Number :	RNC-WPS-016		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW) + Shield Metal Arc Welding (SMAW)		
Type (Manual, Semi-auto, or Automatic) :	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)				POSTWELD HEAT TREATMENT (QW-407)			
Material Spec.	A335	To	A335	Temperature	720° C		
Type or Grade	P11	To	P11	Hold Time	2 HR		
P. No.	4	To	4	Others	Heating Rate 195° C/Hr		
Thickness of Test Coupon	8.18 mm.				Cooling Rate 250° C/Hr		
Diameter of Test Coupon	Ø 8"						
Others				GAS (QW-408)			
				Percent Composition			
					Gas (es)	Mixture	Flow Rate
				Shielding	Argon	99.95%	12 L/Min.
				Trailing	-	-	-
				Backing	-	-	-
FILLER METALS (QW-404)				ELECTRICAL CHARACTERISTICS (QW-409)			
SFA Specification	GTAW			Current	DC (GTAW)		
AWS Classification	A 5.28			Polarity	EN		
Filler Metal F-No.	ER80S-G			Amperage	127 - 154	Voltage	10 - 22
Weld Metal Analysis A-No.	6						
Size of Filler Metal	3						
Others	Ø 2.4 mm.						
Weld Metal Thickness	Kobelco TGS-1CM			Tungsten Electrode Size	Ø 2.4 mm. 2% thoriated.EWTh-2		
				Others	N/A		
POSITION (QW-405)				TECHNIQUE (QW-410)			
Position of Groove	6G			Travel Speed	5 - 7 cm/min.		
Progression of Welding (Uphill or Down)	Up Hill			String or Weave Bead	Both		
Others	N/A			Oscillation	N/A		
PREHEAT (QW-406)				Multiple or single Pass	Multiple Pass		
Preheat Temperature	220° C			Multiple or Single Electrodes	Single Electrodes		
Interpass Temperature (max.)	260° C			Others	N/A		
Others	N/A						

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number: RNC-PQR-015

Tensile Test Report No.TS-13-04-030

Specimen	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (KN)	Ultimate Strength (N/mm ²)	Type of Failure & Loc.
1050/13 TR1	19.50	7.98	155.610	74.923	481.479	OUT OF WELD
1050/13 TR2	19.50	8.18	159.510	77.159	483.725	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-13-04-013

Type and Figure Number	Type of Bend	Indication	Results
1050/13 BF1	Transverse face bend	Open Discontinuity 0.60,0.30 mm	Accepted
1050/13 BF2	Transverse face bend	No Open Discontinuity	Accepted
1050/13 BR1	Transverse root bend	No Open Discontinuity	Accepted
1050/13 BR2	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170) Report No.IM-13-04-016

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Average (Joules)	Lateral Expansion Mils
1050/13-1	Weld Metal	V-Notch	- 20 °C	143.64	113.99	
1050/13-2	Weld Metal	V-Notch	- 20 °C	113.33		
1050/13-3	Weld Metal	V-Notch	- 20 °C	84.99		
1050/13-1	HAZ	V-Notch	- 20 °C	152.12	141.85	
1050/13-2	HAZ	V-Notch	- 20 °C	128.43		
1050/13-3	HAZ	V-Notch	- 20 °C	144.99		
1050/13-1	Base Metal	V-Notch	- 20 °C	143.64	149.74	
1050/13-2	Base Metal	V-Notch	- 20 °C	152.86		
1050/13-3	Base Metal	V-Notch	- 20 °C	152.73		

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A
Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.PQR-003

Hardness Testing Report No. HV-13-04-007

Macrostructure Report No. MA-13-04-015

Welders Name Mr.Tawin M.

Welder Number W-060

Tests Conducted By Mr.Keerati Sa-ngoen

Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor :

Date :

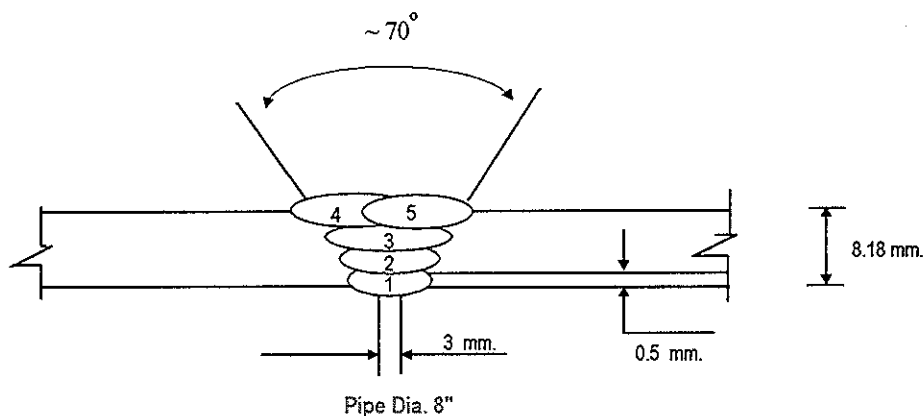
Inspected by : Mr.Wanlop N

Certified by : Mr.Chaloemkiet R

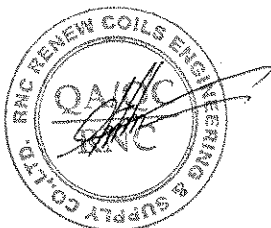
Date : 9 April 2013

PAE TECHNICAL SERVICE	WELDING PROCEDURE	PQR No. : RNC-PQR-016
		WPS No. : RNC-WPS-016
	QUALIFICATION RECORDS	THIS PQR RECORDS ACTUAL TEST CONDITION AND RESULT

Welding Sequences for RNC-PQR-016 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or	Wire Feed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
							Weave Bead	Speed				
1	GTAW	ER80S-G	2.4	127	12	DCEN	-	-	5	18.20	12	-
2	GTAW	ER80S-G	2.4	150	10	DCEN	-	-	5	18.30	12	-
3	SMAW	ER80S-G	2.4	154	10	DCEN	-	-	5	19.40	12	-
4	SMAW	ER80S-G	2.4	127	22	DCEN	-	-	7	23.90	12	-
5	SMAW	ER80S-G	2.4	125	21	DCEN	-	-	7	22.50	12	-



RNC**WELDERS CERTIFICATE**

Record of Welder or Welding Operator Qualification Test



Client : PTT GLOBAL CHEMICAL CO., LTD.

Project Name : PTTGC 2013

Contractor : Renewcoils Engineering & Supply Co., Ltd.

Sub-Contractor : -

Welders Name : Mr. Thawin M.

I.D. No. : 3 55 1000 109 500

Welder No. : W-060

Welding Process (es) used : GTAW

Type : Manual

WPS Number (Used by welder during the welding of test coupon) :

RNC-WPS-016

Base material (s) welded : ASME / ASTM A 335 P 11

Thickness : 8.18 mm. (PQR)



(QW - 350)	Actual Values	Range Qualified
Backing (metal, weld metal, welder from both sides, flux, etc.) (QW-402)	With out Backing	With / With out Backing
ASME P - Number to ASME P- Number (QW - 403)	P-4 + P-4	P - No. 4
Base metal Dimension ☐ Plate ☒ Pipe (include diameter)	OD 2-3/8"	O.D 1" to Unlimited
Filler metal specification (SFA) A 5.28 Classification (QW - 404)	ER 80S-G	F no.6
Filler metal F - Number	F6	F6
Consumable insert for GTAW or PAW	N/A	N/A
Weld deposit thickness for each welding process : GTAW	8.18 mm	Max. 16.36 mm
Welding position (QW - 405)	6G	All Position Unless Downhill
Welding Progression (uphill / downhill) (QW - 405)	Uphill	Uphill
Backing gas for GTAW , PAW , or GMAW ; Fuel gas for OFW (QW-408)	NO	YES or NO
GMAW transfer method / mode (QW - 409)	N / A	N / A
Welding current type and polarity	DC / EN	DC / EN

Visual Inspection Results (QW - 302.4)

Fit - up : Acceptable

Root Pass : Acceptable

Weld Profile : Acceptable

Guided Bend Test Results

Guided Bend Test Type	() QW - 462.2 (Side) Results	() QW - 462.3a (Trans. R&F Type)	() QW - 462.3b (Long. R&F) Results
-	-	-	-
-	-	-	-

Radiographic Test Results (QW - 304 & 305)

ACCEPTABLE SEE REPORT No. PQR-003

(For alternative qualification of groove welds by radiography)

Fillet Weld - Fracture Test Result

-

Length and percent of defects

-

in.

Macro test fusion

-

Fillet leg size

-

in.x

-

in.

Concavity / Convexity

-

in.

Welding test conducted by

PAE TECHNICAL SERVICE CO.,LTD.

Mechanical test conducted by

N / A

Laboratory test No.

N / A

We certify that the statements in this record are correct and that the test coupons were prepared . Welded

and tested in accordance with the requirement of :

ASME - SECTION IX : 2010 / ASME B 31.3 : 2010

	Prepared By	Reviewed By / Approved By	Approved By
Signature :			
Name :	Mr. Chatchawan S.		
Date :	11/04/13.		

PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)682208-9, Fax : 038)682206, SRIRACHA : 038)330220, 330229

Page : 1 Of 1[illegible]



PAE TECHNICAL SERVICE CO.,LTD
NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT
 69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250
 Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 682208-9

WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 1

Client : Renew Coils Engineering & Supply Co.,Ltd.		Project Name : MINI SHUTDOWN PTTGC		Location : RNC Work Shop		Report No. PQR-003	
		Customer Order No. : N/A		WPS No. : RNC-WPS-016		Date : 29-Mar-13	

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
W-060	MR.TAWIN M.	GTAW	6G	ASME IX	A335 P11	ø 8" Thk. 8.18 mm.	ER80S-G	ACCEPTED	ACCEPTED

Joint Details

Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	GTAW	ER80S-G	2.4	DCEN	127	12	5	224	18.2	Argon	12	N/A	N/A
2	GTAW	ER80S-G	2.4	DCEN	150	10.2	5	213	18.3	Argon	12	N/A	N/A
3	GTAW	ER80S-G	2.4	DCEN	154	10.5	5	221	19.4	Argon	12	N/A	N/A
4	GTAW	ER80S-G	2.4	DCEN	127	22	7	220	23.9	Argon	12	N/A	N/A
5	GTAW	ER80S-G	2.4	DCEN	125	21	7	210	22.5	Argon	12	N/A	N/A

PAE Inspection By
 SIGNATURE :
 NAME : Mr. Wanlop N.
 DATE : 30-Mar-2013

Witness By
 SIGNATURE :
 NAME :
 DATE : 31/04/13

Approved By
 SIGNATURE :
 NAME : ภูชิน สีหอมไชย
 DATE : Phuchin Seehornchai



GLOBAL HEAT TREATMENT (THAILAND) CO.,LTD.

PREHEAT REPORT

CLIENT:	RENEW COILS ENGINEERING & SUPPLY CO.,LTD.		
PROJECT:	MINI SHUT DOWN PTTGC		
CONTRACTOR:	GLOBAL HEAT TREATMENT (THAILAND) CO.,LTD.		
PREHEAT METHOD:	☐ GAS	☒ ELECTRIC	COMPLETION DATE : 29-03-2013
LOCATION:	☒ SHOP	☐ FIELD	REPORT NO. PH -001
RECORDER NO:	EH- 07ZA194		PAGE NO. 2/2
CHART SPEED:	☒ 25 MM/Hr	☐ 50 MM/Hr	☐ 100 MM/Hr

RESULT RECORDER CHART TO BE ATTACHED

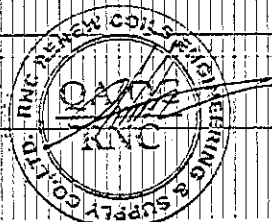
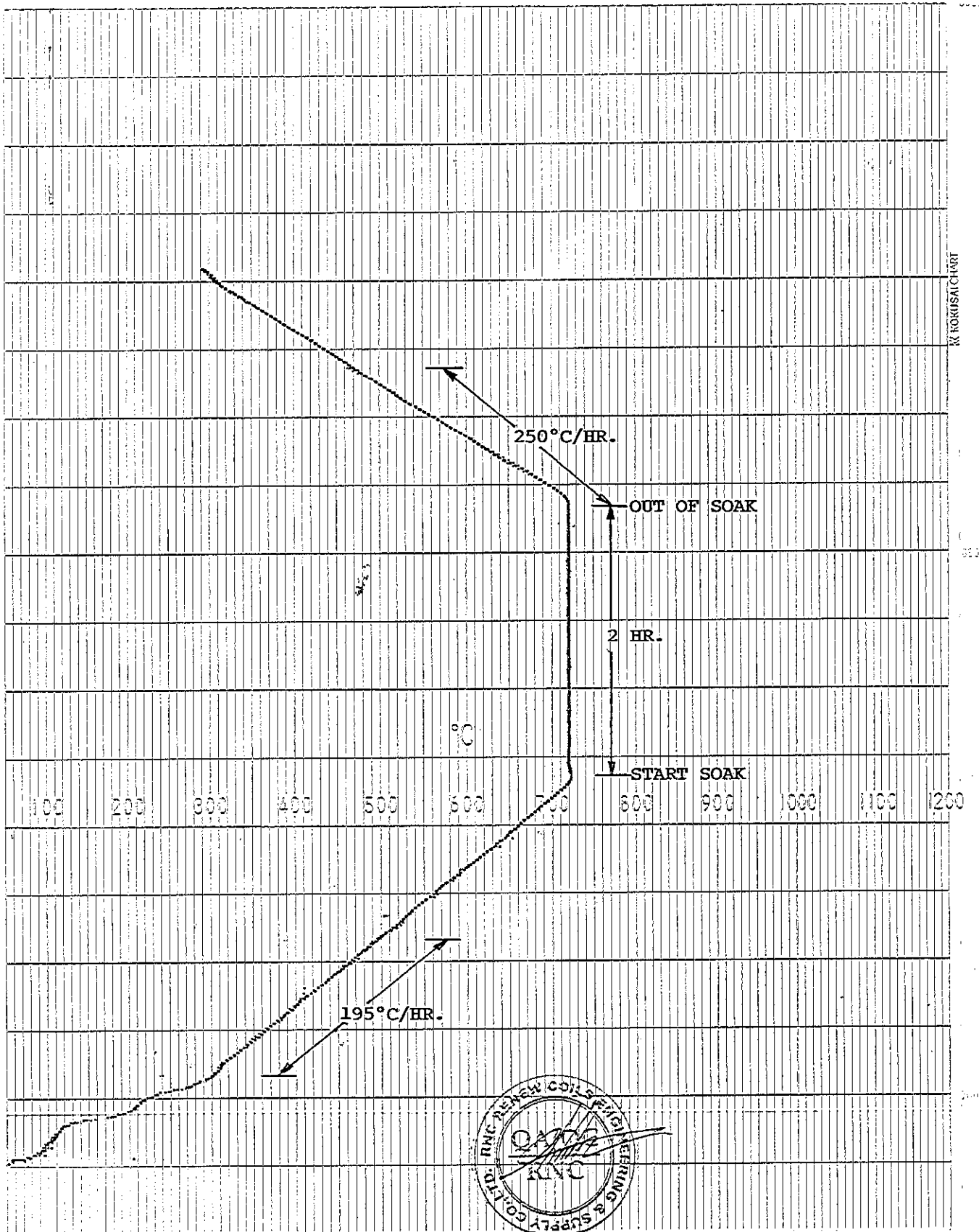
DESCRIPTION	REQUEST	ACTUAL
1) PREHEAT TEMP (C° MIN)	200-250	220
2) INTERPASS TEMP (C° MAX)	300	260
3) POSTHEAT TEMP (C°)	-	-
4) POSTHEAT TIME (Hr)	-	-

MATERIAL : **A335 P11**

[illegible]

TOTAL	2	JOINTS.
TOTAL	16	DB.

CHECKED BY GLOBAL HEAT TREATMENT (THAILAND)	APPROVED BY	APPROVED BY
SIGN: 	SIGN: 	SIGN: 
NAME: ROMKHUN D.	NAME: Mr. Watcharam	NAME: ภูชิน สีห้อมไชย
DATE: 29-03-2013	DATE: 29/3/13	DATE: Phuchin Seehomcha



W. KOKUSAI CHART

353

354

DATE. 30-03-2013

CHART NO. PW - 001

RECORDER NO. EH 07ZA194

CHART SPEED. 25mm./HR.

CUSTOMER. RENEW COILS ENGINEERING & SUPPLY CO., LTD.

PROJECT. MINI SHUT DOW PITGC

PQR NO. RNC-PQR-015

SIZE. 8" THK. 8.18 mm.

PQR NO. RNC-PQR-016

SIZE. 8" THK. 8.18 mm.

MATERIAL. A335 P11

START TIME. 10:00 AM.

TECHNICIAN. MR. CHAINIKORN N.

TC	COLOUR
1.	RED
2.	BLACK
3.	SKY BLUE
4.	GREEN

CHART NO. ZT 201

KOKUSAI CHART



GLOBAL HEAT TREATMENT (THAILAND) CO.,LTD.

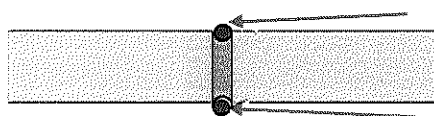
HEAT TREATMENT REPORT

CLIENT:	RENEW COILS ENGINEERING & SUPPLY CO.,LTD.		
PROJECT:	MINI SHUT DOWN PTTGC		
CONTRACTOR:	GLOBAL HEAT TREATMENT (THAILAND) CO.,LTD.		
DESCRIPTION:	☒ PWHT ☐ NORMALIZE ☐ ANNEALING ☐ STABILIZE		
LOCATION:	☒ SHOP ☐ FIELD	REPORT NO.	PW-001
RECORDER NO:	EH- 07ZA194	PAGE NO.	1
CHART SPEED:	☒ 25 MM/Hr ☐ 50 MM/Hr ☐ 100 MM/Hr		
METHOD BY:	☒ ELECTRIC ☐ GAS ☐ OIL ☒ LOCAL ☐ FURNACE ☐ INTERNAL		
COMPLETION DATE:	30-03-2013	MATERIAL:	A335 P11

HEAT TREATMENT CONDITION / RESULTS

DISCRIPTION	REQUEST	ACTUAL	
1) LOADING TEMP (C°)	300	300	
2) HEATING RATE (C°/Hr)	220	195	
3) HOLDING TEMP (MIN-MAX)	704-746	720	
4) HOLDING TIME (Hr)	2 HR.	2 HR.	
5) COOLING RATE (C/Hr)	250	250	
6) UNLOADING TEMP (C°)	300	300	

LINE NO.& CLASS & SHT NO.& SPL.NO. DWG.NO & TANK NO.	WELD NO.	SIZE	THK. (MM)	WELD TYPE	T/C NO.
PQR NO. RNC-PQR-015		8"	8.18	BW	1,2
PQR NO. RNC-PQR-016		8"	8.18	BW	3,4



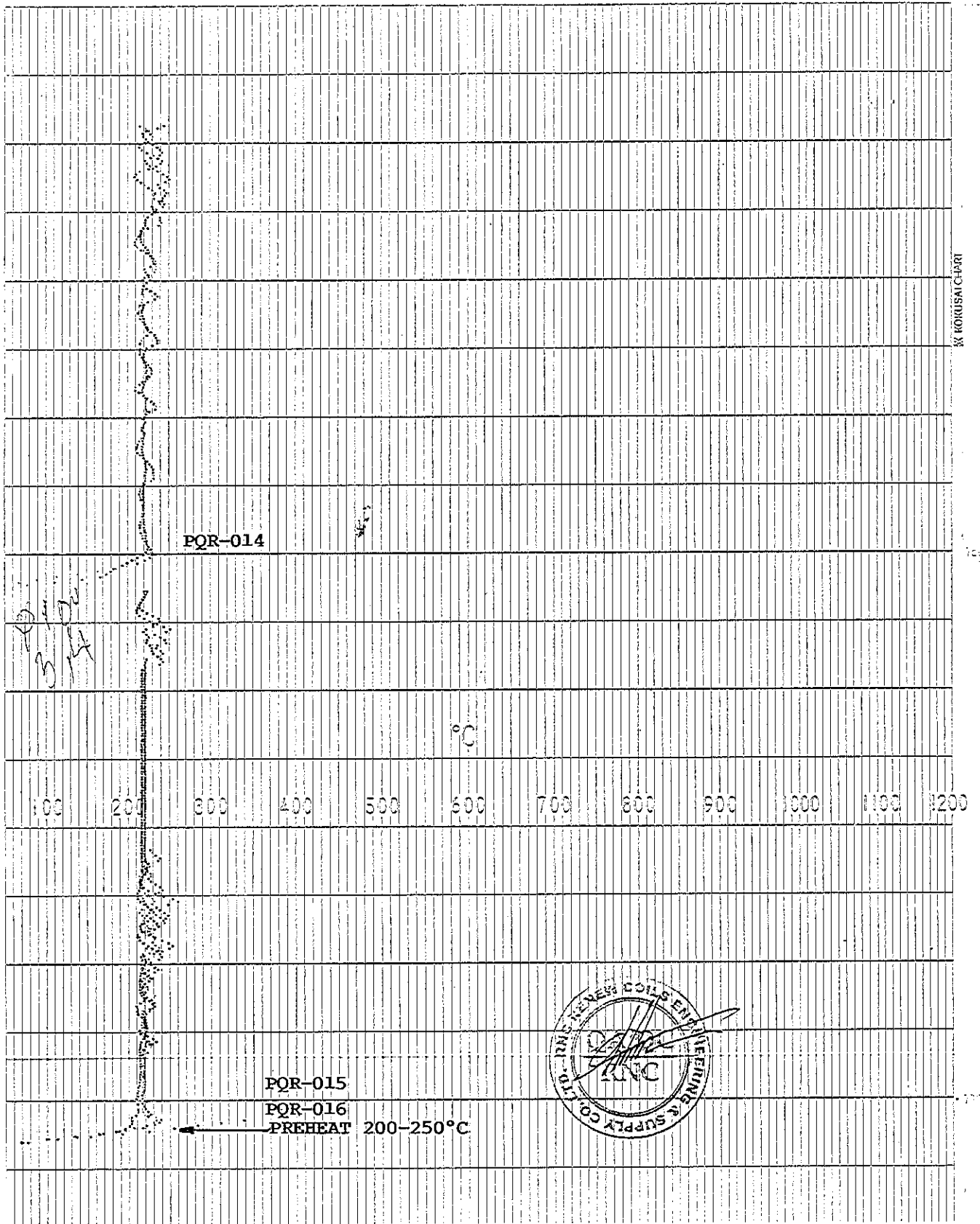
T/C NO.

T/C NO.

TOTAL 2 JOINTS.

TOTAL 16 DB.

CHECKED BY GLOBAL HEAT TREATMENT (THAILAND)	APPROVED BY	APPROVED BY
SIGN:	SIGN:	SIGN:
NAME: KOMKHUN D.	NAME: Mr. Chatchanan S.	NAME: ภูชิน สีหอมไชย
DATE: 30-03-2013	DATE: 10/4/13	DATE: Phuchin Seehomchai



15	DATE 29-03-2013										TC	COLOUR	
14	CHART NO. PH - 001										1.	RED	
											2.	BLACK	
											3.	SKY BLUE	
											4.	GREEN	
14	RECORDER NO. EH 07ZA194												
	CHART SPEED 25mm./HR.												
13	CUSTOMER. RENEW COILS ENGINEERING & SUPPLY CO.,LTD.												
	PROJECT. MINI SHUT DOW PITGC												
	PQR NO. RNC-PQR-014												
12	SIZE. 10"	THK. 9.27 mm.	400	500	600	700	800	900	1000	1100	1200		
	PQR NO. RNC-PQR-015												
	SIZE. 8" THK. 8.18 mm.												
11	PQR NO. RNC-PQR-016												
	SIZE. 8" THK. 8.18 mm.												
	MATERIAL. A335 P5, A335 P11												
10	START TIME. 09.30 AM.												
	TECHNICIAN. MR. ANUSORN R.												

CHART NO. ZT 201

K KONGSACHART



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnmp.co.th



8943

TESTING
NA 0101

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-13-04-030

Revision No. 0

Test Product	WPS No. RNC-WPS-016	Material Characterization	Pipe
Project Name	MINI SHUTDOWN PTTGC	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A335 P11	Type of Test	Reduced Section Tension
Dimension	Ø 8" (8.18 mm Thk.)	Equipment's Capacity	200 TONS
Date Received	03 April 2013	Equipment / Serial No.	SHIMADZU UH200A / 32098926
Date Tested	05 April 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	1050/13 TR1	N/A	7.98	19.50	155.610	N/A	N/A	N/A	74.923	481.479	N/A	OUT OF WELD
2	1050/13 TR2	N/A	8.18	19.50	159.510	N/A	N/A	N/A	77.159	483.725	N/A	OUT OF WELD

The unit of N/mm² is equivalent to MPa

Additional details : Welding Process / Position : GTAW / 6G

Welder Name : MR.TAWIN M.

Welder No. : W-060

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

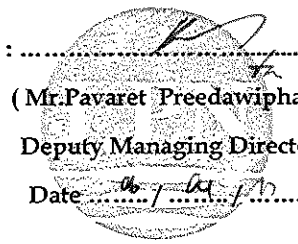
Date/...../.....

Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date/...../.....

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmpm.co.th

28946

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-13-04-013

Revision No. 0

Test Product	WPS No. RNC-WPS-016	Material Characterization	Pipe
Project Name	MINI SHUTDOWN PTTGC	Test Method	ASME Sec. IX : 2010 edition
Project No.	N/A	Standart Of Test	ASME Sec. IX : 2010 edition
Material Specification	A335 P11	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 8" (8.18 mm Thk.)	Diameter of Meudrel	32 mm
Date Received	03 April 2013	Bend Angle	180 Degree.
Date Tested	05 April 2013	Equipment/ Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	1050/13 BF1	Transverse face bend	8.16x37.89x160.74 mm	Open Discontinuity	WELD ZONE	0.60, 0.30 mm
2	1050/13 BF2	Transverse face bend	8.20x37.89x161.59 mm	No Open Discontinuity	N/A	-
3	1050/13 BR1	Transverse root bend	8.07x37.90x161.38 mm	No Open Discontinuity	N/A	-
4	1050/13 BR2	Transverse root bend	7.94x37.90x161.83 mm	No Open Discontinuity	N/A	-

Additional details : Welding Process / Position : GTAW / 6G

Welder Name : MR.TAWIN M.

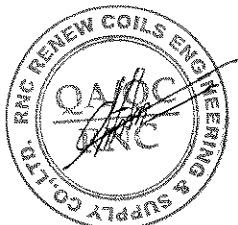
Welder No. : W-060

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date / /

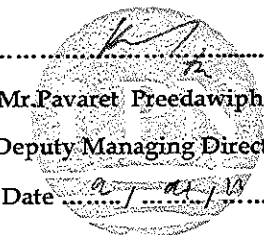


Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date / /

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8949

TESTING
No. 0136

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-13-04-016

Revision No. 0

Test Product	WPS No. RNC-WPS-016	Material Characterization	Pipe
Project Name	MINI SHUTDOWN PITGC	Test Method	ASTM A370-10 and ASTM E23M-07a ^{e1}
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A335 P11	Type of Test	Charpy V-Notch Impact
Dimension	Ø 8" (8.18 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	03 April 2013	Equipment 's Capacity	542 Joules
Date Tested	05 April 2013	Ambient Temperature	23 ± 5 °C

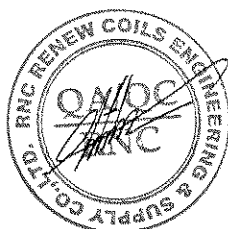
Item No.	Test No.	Test Temperature (°C)	Specimen Dimension			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	1050/13-1	-20°C	5.001	10.009	54.88	143.64	113.99	Weld Metal	-
2	1050/13-2	-20°C	5.000	10.009	54.85	113.33			
3	1050/13-3	-20°C	4.998	10.008	54.86	84.99			
4	1050/13-1	-20°C	5.000	10.006	54.80	152.12	141.85	HAZ	-
5	1050/13-2	-20°C	5.000	10.013	54.76	128.43			
6	1050/13-3	-20°C	5.000	10.008	54.75	144.99			
7	1050/13-1	-20°C	5.000	10.009	54.89	143.64	149.74	Base Metal	-
8	1050/13-2	-20°C	5.003	10.008	54.90	152.86			
9	1050/13-3	-20°C	4.998	10.009	54.88	152.73			

Additional details : Welding Process / Position : GTAW / 6G

Welder Name : MR.TAWIN M.

Welder No. : W-060

Reported by : 
 (Mr.Keerati Sa-ngoen)
 (Acting) General Manager - Quality Assurance
 Date 06 / 04 / 13



Approved by : 
 (Mr.Pavaret Preedawiphat)
 Deputy Managing Director
 Date 06 / 04 / 13

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Macrostructure Analysis Report

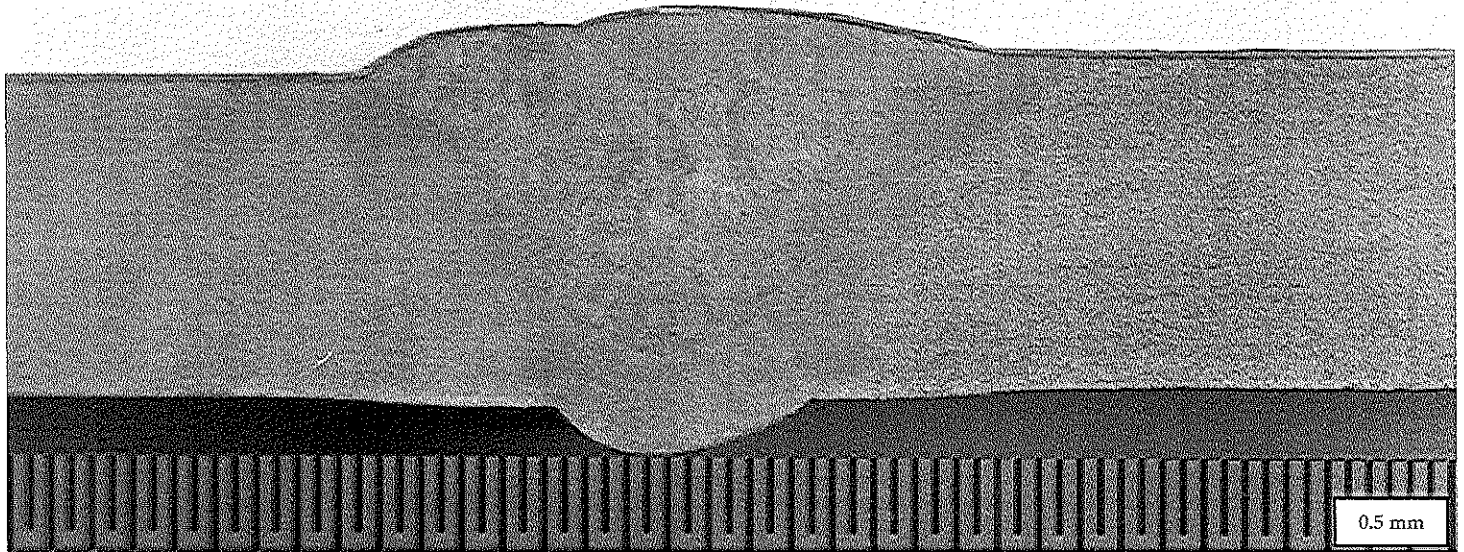
Page 1 of 1

Report No. MA-13-04-015

Revision No.0

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.**Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000 Thailand.**

Test Product	WPS No. RNC-WPS-016	Material Characterization	Pipe
Test No.	1050/13 MA	Equipment used	Microscope upto 40X (times)
Test Location	-	Serial No.	LEICA 0650854912
Material Specification	A335 P11	Method of Preparation	ASME Section IX : 2010 edition
Dimension	Ø 8" (8.18 mm Thk.)	Test Method	ASME Section IX : 2010 edition
Welding Process/Position	GTAW+SMAW/6G	Standard of Test	ASME Section IX : 2010 edition
Type of Joint	Butt Joint	Etchant Reagent	25 ml HNO ₃ + 75 ml H ₂ O
Welder Name / No.	Mr. Tawin M. / W-060	Time of Etching	5 minutes
Date Received	3 April 2013	Date Tested	6 April 2013

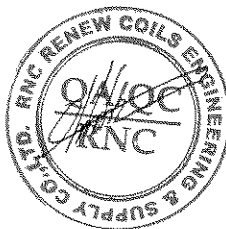
**Macro-etch Examination :** The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.**Additional details :** Project Name : MINI SHUTDOWN PTTGC

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date : 09/04/13



Approved by :

(Mr.Payaret Freedawaphat)

Deputy Managing Director

Date : 09/04/13

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13293

Hardness Testing Report

Page 1 of 1

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

Report No. HV-13-04-007

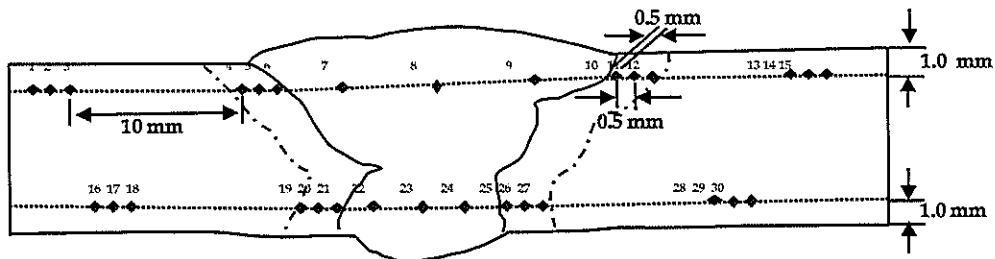
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000 Thailand.

Revision No. 0

Test Product	WPS No. RNC-WPS-016	Material Characterization	Pipe
Test No.	1050/13 HN	Test Method	ASTM E92-82 (Reapproved 2003) ^{E2}
Project Name	MINI SHUTDOWN PTTGC	Standard of	ASME Section IX : 2010 edition
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A335 P11	Equipment	Wilson Wolpert , 432 SVD (V32D219)
Dimension	Ø 8" (8.18 mm Thk.)	Indenter Size	Pyramid diamond, angle of 136°
Welding Process/Position	GTAW+SMAW/6G	Test Load	HV 10 Kgf.
Type of Joint	Butt Joint	Total Force Dwell Time	15 s
Date Received	3 April 2013	Cal. Block	No. A60023 No. A62072
Date Tested	8 April 2013	Ambient Temperature	25.6 °C

Test Location

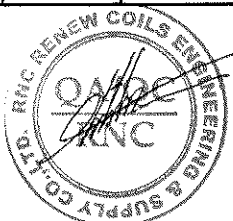
Total = 30 points



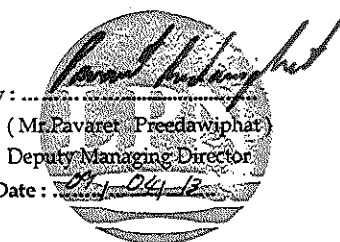
Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale)	Point No.	Location	Hardness value (Scale)
1	BASE	140.9	16	BASE	141.8	*****		
2	BASE	140.3	17	BASE	139.5			
3	BASE	143.3	18	BASE	139.2			
4	HAZ	151.6	19	HAZ	146.4			
5	HAZ	168.5	20	HAZ	155.5			
6	HAZ	181.9	21	HAZ	161.0			
7	WELD	185.2	22	WELD	166.5			
8	WELD	196.5	23	WELD	166.2			
9	WELD	190.6	24	WELD	165.8			
10	HAZ	190.2	25	HAZ	158.0			
11	HAZ	187.0	26	HAZ	151.2			
12	HAZ	169.8	27	HAZ	153.0			
13	BASE	138.9	28	BASE	141.1			
14	BASE	137.8	29	BASE	139.8			
15	BASE	138.1	30	BASE	139.6			

Additional details : Welder Name: Mr. Tawin M. / W-060

Reported by : *Parat Preechawiphat*
 (Mr. Keerati Sa-ngoen)
 (Acting) General Manager - Quality Assurance
 Date : 09/04/13



Approved by : *Parat Preechawiphat*
 (Mr. Parat Preechawiphat)
 Deputy Managing Director
 Date : 09/04/13



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SIGNATURE :	PAE INSPECTED BY	REVIEWED / WITNESSED BY	APPROVAL BY
		 DA/OC	
	NAME : Mr. Anusorn M.	Mr. Chatchawan	Phuchin Seemachai
DATE :	10-Apr-13	11/04/13	

7

ER 809-G

EX-808-9.

B.52743

PMT

*Matilwam PQR. P44 13008701501

Tenaris

INSPECTION CERTIFICATE
CERTIFICATO DI COLLAUDO
(UNI EN 10204 3.1 / ISO 10474 3.1 B)

Number / Numero: **11/04825**
Page / Pagina: **1/8**
Date / Data: **March 11, 2011**

Deffina S.p.A. - Via F.lli Pelli
Piazza Cesare G. Luigi 1846, 1
20124 Bergamo (BG) - Italy
Tel 035 580 13110
Fax 035 580 5527 fax

Customer / Cliente: VD004131080000273		Customer's Order Ref. / Rifer. Ordine Cliente - Item: 1210324/d17	
Manufacturer Process / Processo di fabbricazione: NORMALIZED AND TEMPERED NORMALIZZATI E RINVENUTI		Product Type / Tipo di prodotto: See note nr.1 Vedi nota nr.1	
Standard or Specification / Norme o specifiche: See note nr.3 Vedi nota nr.3		Steel Grade / Grado dell'acciaio: STEEL GR. P 11 ASTM A 335 AND ASME SA 335 ACCIAIO GR. P 11 ASTM A 335 E ASME SA 335	
Dimensions / Dimensioni: ø 219.1mm O.D. x 8.18mm W.T.	Schedule / Scheda: 10000 mm ± 11800 mm	Quantity / Quantità: 1911095 ft 582.7 mt	Weight / Peso: 85648.8 lb 25241 kg

DELIVERY NOTES / AVVISI DI SPEDIZIONE

Job number: 11503045/18	Delivery Notes / Avvisi di spedizione
Shipping note: 00008850 - 11/03/2011	
Address: VIALE LIPICE 40	
	Town: 10095 GRUGLIASCO-TO
	Country: ITALIA

TENSILE TEST / PROVA DI TRAZIONE

Heat N° Colata N°	Sample N° Prova N°	Lot N° Lotto N°	Specimen condition Condizioni della provetta				Specimen dimensions Dimensioni della provetta		Test Temp. Temp. di prova	Y.S.		U.T.S.		Elongation / Allungamento			
			Lt	Se	Type	Ort	ASPA	Area		EUL 0.5%	Reg	Min	Max	Lo	Min	Max	Obt.
910659	R9613	1	B	NT	SS	L	38.00 x 7.00	301.90	120	386.00	517.00	25.00	50.00	25.00	29.8%	46.0%	
910659	R9614/C	1	B	NT	SS	L	38.00 x 7.00	301.90	120	338.00	510.00	25.00	50.00	25.00	29.8%	46.0%	
910659	R9615/C	1	B	NT	SS	L	38.10 x 7.00	302.50	120	389.00	518.00	25.00	50.00	25.00	29.8%	46.0%	
910659	R9616/C	1	B	NT	SS	L	38.10 x 8.00	306.40	120	340.00	497.00	25.00	50.00	25.00	30.0%	44.0%	
910659	R9617	1	B	NT	SS	L	38.00 x 8.20	312.40	120	314.00	480.00	25.00	50.00	25.00	30.3%	46.0%	
910659	R9618/C	1	B	NT	SS	L	38.20 x 8.70	323.10	120	367.00	491.00	25.00	50.00	25.00	30.0%	43.0%	
910659	R9619	1	B	NT	SS	L	38.00 x 8.40	321.00	120	324.00	509.00	25.00	50.00	25.00	30.7%	46.0%	



Body / Corpo	Orientation / Orientamento	Orientation / Orientamento	Orientation / Orientamento
Longitudinal / Longitudinale	Normalized and tempered / Normalizzato e rinvenuto	Required / Richiesto	U.T.S. Ultimate Tensile Strength / Resistenza a Trazione
Longitudinal / Longitudinale			Y.S. Yield Strength / Snervamento

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FOR03171

271322/JVD

Your order n°: **IP236915-18967**
Our order n°: **03738012/PL It: 1A**
Adv. of Dispmr: **439280**
Description: **PIPE PIPES DRL 8 STD P11**
Heat n°: **910659**
Qty: **116.50**
JORDI VAN BAARLE
Blz 1: 8